

Xiaohan Cai

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Research Interests

Differential Geometry and Partial Differential Equations

Education

Ph.D. in Mathematics

Shanghai Jiao Tong University

2022–Present

Shanghai, China

- Advisor: Prof. [Congming Li](#), Prof. [Mijia Lai](#)

- Dissertation title:

B.Sc. in Mathematics

Shanghai Jiao Tong University

2018–2022

Shanghai, China

Visiting student

BICMR and Peking University

Spring 2021

Beijing, China

Visiting positions

Visiting scholar

Michigan State University

December 2024–May 2026

Host: Prof. [Xiaodong Wang](#)

Worked on uniqueness for solutions to geometric PDEs on compact manifolds.

Publications and Preprints

1. *Liouville equations on complete surfaces with nonnegative Gauss curvature* (with Mijia Lai)
Pacific J. Math. 332 (2024), no. 1, 23–37. [Journal link](#), [arXiv:2309.01956](#).
2. *A survey of classification results for solutions to the Liouville equation* (with Mijia Lai).
J. Math. Study 57 (2024), no. 3, 398–424. [Journal link](#) .
3. *An area growth estimate of the Liouville equation* (with Mijia Lai and Chilin Zhang).
Accepted by Math. Res. Lett. [arXiv:2409.19327](#).
4. *Uniqueness results for positive harmonic functions on manifolds with nonnegative Ricci curvature and strictly convex boundary*.
Calc. Var. Partial Differential Equations 65 (2026), no. 6, Paper No. 193, 23 pp.
[Journal link](#), [arXiv: 2511.09994](#).
5. *A modified Bakry-Émery Γ_2 criterion inequality and the monotonicity of the Tsallis entropy*
(with Xiaodong Wang). [arXiv:2512.15525](#).

Seminar and Conference Talks

- Workshop on geometric analysis SJTU, Shanghai, China
Scalar curvature integration on conformally flat manifolds with nonnegative Ricci curvature
May 15-17, 2026

- The 7th National PhD Student Conference on Partial Differential Equations SZU, Shenzhen, China
Some classification results for solutions to Liouville equations October 25-27, 2024
- NJU-SJTU geometric analysis seminar NJU, Nanjing, China
Liouville equations on complete surfaces with nonnegative Gauss curvature October 10-23, 2023
- International Conference on PDEs and Geometric Analysis SJTU, Shanghai, China
Asymptotic behavior of solutions to some semilinear equations on $\mathbb{R}^2$ June 19-23, 2023
- Graduate Student Forum on Spectral Geometry USTC, Anhui, China
Volume comparison from the viewpoint of isoperimetric problem March 17-20, 2023

Teaching

Teaching assistant at Shanghai Jiao Tong University:

- Real Analysis Spring 2022
- Differential Geometry Fall 2021

Honors and Awards

- Zhu Gongjin Scholarship, Shanghai Jiao Tong University 2025
- Zhiyuan Outstanding Student Scholarship, Shanghai Jiao Tong University 2022
- Shanghai Outstanding Undergraduate 2022
- Zhiyuan Honors Scholarship, Shanghai Jiao Tong University 2018-2022

Skills

Languages Mandarin Chinese (native), English.